Solar Filament Instance Segmentation from GONG H-alpha Images: Consensus Models, Morphology-Aware Refinement and Evidence-Based Selection

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Verified best official submission: Kaggle ID **58112557**, publicly scored **PQ = 0.39**, submitted 8 November 2026.

1. Abstract

We study reproducible instance segmentation of thin solar filaments in 2048 × 2048 GONG H-alpha full-disc images. Our method combines independently trained convolutional segmenters, an object-proposal branch, instance-wise selection and a pair of native-image crop refiners. Every inference feature comes from the supplied test image, never from test labels, dates, observatory identities, adjacent frames or externally published annotations. Evaluation follows the challenge's Panoptic Quality (PQ) definition, pooling true positives, false positives, false negatives and intersection-over-union over **individual annotator-image records**, while resampling uncertainty over physical observations.

Four temporal-block experiments, deliberately purged around held-out dates, gave our H6 control local PQ 0.4465; the two-refiner H9b variant reached 0.4502 in the exploratory cross-block analysis, but its improvement did not meet our pre-registered promotion threshold. The pixel-only H9b no-observatory submission nevertheless achieved verified public PQ **0.39**. This report distinguishes public measurements from local validation and preserves unfavorable results, essential to interpreting a small partially labeled leaderboard.

2. Task, data and scoring

MAGFiLO consists of **707 labeled training observations** and **180 competition test images**. The 707 training images contain **1,154 annotation records** and **8,199 annotated filament instances**. Filaments are sparse: median instance area 1,228 native pixels, and a substantial tail consists of structures narrower than ten pixels. Individual physical observations may be annotated independently by two or three experts, creating genuine morphological ambiguity.

For masks accepted at intersection-over-union (IoU) strictly above 0.5, the competition scores

$$\mathrm{PQ} = \frac{\sum_{(g,p)\in TP} \mathrm{IoU}(g,p)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}.$$

Predictions are scored separately against each human annotation record for an observation, before aggregating counts. Matching, not image-wise averaged Dice, determines the reported PQ. Our native 2048-pixel evaluator and independent parity checks reproduce that record-wise interpretation.

We group all annotators of the same physical image before splitting and use **four temporal blocks**, each with a five-day exclusion buffer between training and held-out observations. This harder-than-random split reduces near-duplicate leakage. A calibration subset selects thresholds and post-processing; reserved blocks provide paired comparisons. Our bootstrap confidence intervals draw physical images, preserving within-image annotator dependence.

3. Prediction system

Image input and segmentation. Grayscale H-alpha observations are normalized by disc-local statistics and processed at native 2048 × 2048 resolution, with 512-pixel crops used for network training. We train a ResNet34-encoder U-Net on a soft annotator-consensus foreground target. A complementary EfficientNetV2-S U-Net provides a different representation of thin structures. Independent models are trained on temporally separated subsets, with reproducible seeds and frozen weights.

Instance hypotheses. A Mask R-CNN proposal branch supplements semantic maps. Candidate foreground components are separated into exclusive instance masks. Probability fusion, minimum area, peak/mean confidence and a learned keep rule are fitted without consulting the public leaderboard. The keep decision follows a marginal-PQ principle: an extra proposal is valuable only when its expected overlap benefit exceeds its expected false-positive cost. For example, at PQ near 0.39 and mean matched IoU near 0.64, an added instance needs approximately a 0.30 match probability to break even.

Native-detail refinement. H9b evaluates a matched pair of independently initialized crop refiners, one built around ResNet34 and another around EfficientNetV2-S. Each receives an image-centred crop with the proposal seed and predicts a sharper instance mask, retaining thin barbs and curled filament ends. Model scores are used to refine candidate geometry, not to infer any unavailable label. The *no-site* variant explicitly excludes observatory identity and date metadata; during test inference it processes **only the single supplied H-alpha image**. This exclusion also rules out temporal-neighbor test-time augmentation.

Disentangling research variants. Mask R-CNN, U-Net consensus, NeuroPixel and per-instance refiners were evaluated as separately attributable interventions. The distinct experimental P58 small-object sampler and P59 train-only reliability gate, studied after the leaderboard entry, remain non-promoted and are not components of the 0.39 submission.

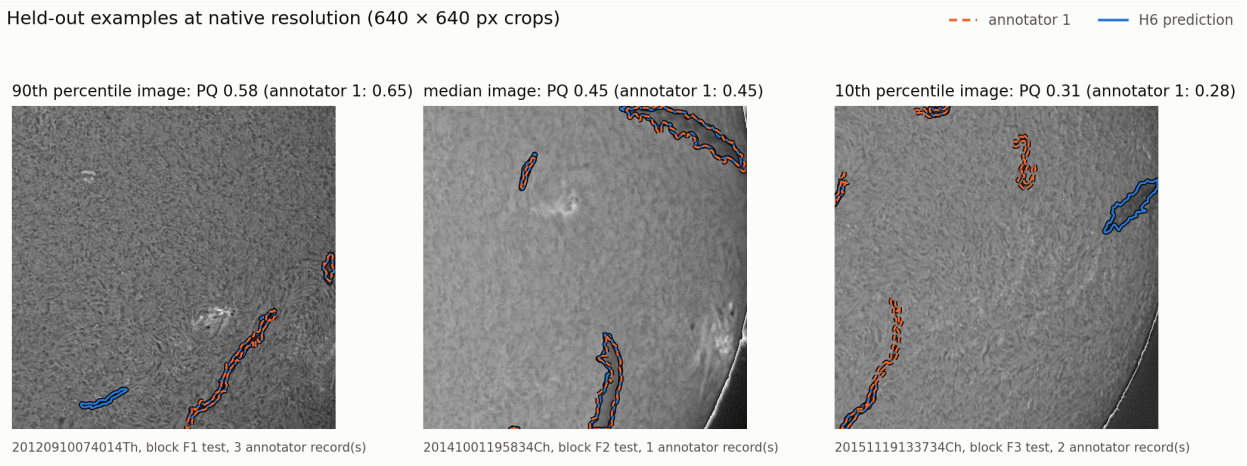


Figure 1: Audited H6 examples: high-, median- and low-quality filament predictions, shown against original solar observations.

Reproducibility contract. Model selection is based on train/calibration partitions rather than iterative public score probing. Final inference requires fixed checkpoint hashes, exact preprocessing, the original 2048-pixel mask geometry and COCO RLE encoded in Fortran order. One submission row corresponds to one disjoint predicted filament; duplicate masks and inherited labels from externally available test frames are prohibited.

4. Quantitative evidence

The official score is a **public-leaderboard value** rounded to two decimal places on approximately half the test observations; internal validation uses independent, labeled training splits. They are not directly interchangeable.

Candidate / official submission	Public PQ	Evidence
Four-block Mask R-CNN control (P1)	0.32	6 Nov 2026
NeuroPixel multiscale ensemble	0.35	28 Sep 2026
Three-family P9	0.36	6 Nov 2026
Three-family P12	0.37	7 Nov 2026
H6 U-Net/Mask R-CNN learned selector	0.38	7 Nov 2026
H9b pixel-only two-refiner submission	**0.39**	**8 Nov 2026; ID 58112557**

Paired temporal validation. H6 obtained local native PQ **0.4465** over 567 purged observations. H9 achieved **0.4491** (delta +0.0026, 95% interval [+0.0002, +0.0051], positive in three of four blocks). The two-refiner H9b reached **0.4502** (delta +0.0037, 95% interval [+0.0010, +0.0064], positive in four blocks). The variant without observatory features yielded delta **+0.0030** [approximately +0.00001, +0.0059], positive in three blocks. Our frozen promotion rule required a positive delta in at least three blocks, an average paired delta above +0.005 and a strictly positive 95% interval. Thus **H9b is exploratory, not a gate-passing cross-block improvement**, even though it scored better publicly.

Morphology and failure modes. H6 retained-set recall was **0.622** over 6,327 reference filaments. Recall by native area was 0.379 under 500 pixels, 0.632 at 500–1,000, 0.686 at 1,000–2,500, 0.685 at 2,500–7,000 and 0.594 above 7,000. For matched objects, median IoU was **0.689**; per-record Dice median was **0.700**. Approximately 11.2% of reference instances were near misses in IoU [0.3, 0.5), indicating that boundary refinement and small-filament recall are meaningful directions, but they must not be selected on the hidden test.

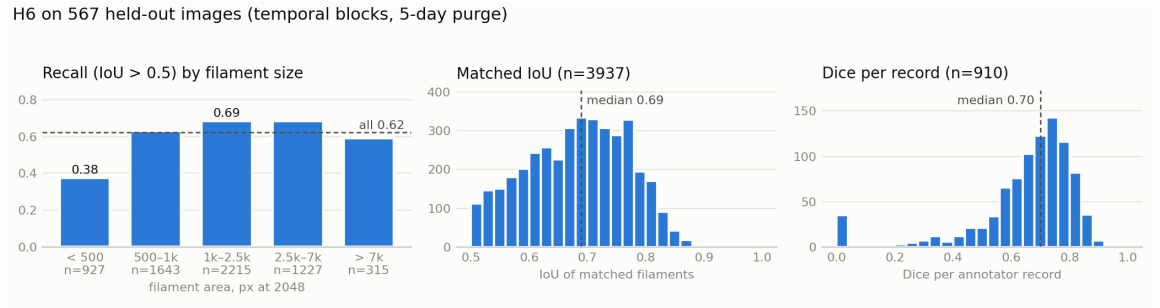


Figure 2: H6 sensitivity to filament area and matched overlap on held-out training observations.

5. Negative controls and falsifiable research

Unsuccessful interventions are preserved to limit selective reporting. A Mask R-CNN-only public baseline scored 0.32, despite stronger results in some training partitions. A deterministic dark-ridge segmentation system scored only 0.087–0.103 in temporal blocks. A low confidence cutoff produced 94 additional true positives but 278 additional false positives on F4, consistent with the high PQ cost of unsupported proposals.

P58, 12 November 2026: A paired small-filament crop study, using **446 F1-train images**, completed 3,600 optimization steps per arm on a single RTX 3090. Its fixed eight-view F1 calibration measured 117 observations and 190 annotator records. The treatment exceeded its paired control (PQ **0.370934** vs **0.359912**; delta **+0.01102**, image-bootstrap interval [+0.00481,+0.01682]), but remained far below the frozen H9b baseline **0.462433** (delta -0.09150), mainly because of false positives (1,770 vs 505). Consequently, it was not promoted or submitted.

P59, 12 November 2026: An eight-feature reliability classifier was fitted only on predictions for those 446 training images (4,651 predicted components, 2,219 positive matches). Its pre-registered small-instance area, overlap and confidence filters admitted **zero** of 1,893 calibration components. H9b baseline PQ remained **0.462433**; delta **0.00000**, a documented **HOLD**. These studies demonstrate why a favorable within-family result is insufficient when the stronger reference model is unchanged.

6. Reproducibility, efficiency and limitations

All experimental checkpoints, source hashes, seeds, manifests and held-out metrics are retained. The best 0.39 Kaggle entry is independently identifiable as **submission 58112557**, named `submission_h9b_nosite.csv`. Our experimental reports specify training-only data, fixed image-level splits and code provenance. The GPU0 paired P58 training recorded **970.34 s for 3,600 paired steps**, peak CUDA allocation **1.793 GiB**, on a 24-GiB RTX 3090. This figure is **training throughput**, not end-to-end inference latency, and therefore must not be misreported as the challenge's efficiency criterion.

The central limitations are annotation disagreement, temporal correlation, small objects, a partial and rounded public leaderboard, repeated historical use of some local folds, and the fact that the report's latest P58/P59 calibrations cover F1 only. We make no hidden-test performance claim. Reproducibility for the final competition submission additionally requires the public Git repository to include **all necessary weights**, pinned `requirements.txt`, an executable Jupyter notebook, licensed dependencies and a consistent 2048 × 2048 COCO-RLE output. Completion of that repository and the organizers' Google form must be independently verified; a public leaderboard score alone does not qualify a team for final judging.

References

[1] A. Ahmadzadeh *et al.*, *Solar Filament Segmentation Challenge 2026*, Kaggle, 2026.
[2] A. Kirillov *et al.*, "Panoptic Segmentation," CVPR, 2019, doi:10.1109/CVPR.2019.00963.
[3] MAGFiLO dataset, *Scientific Data*, 2024, doi:10.1038/s41597-024-03876-y.
[4] Kaggle challenge self-evaluation notebook, updated August 2026, by competition organizers.

Competition code provenance: Agnuxo1/solar-filament-2026-dev (development repository; final release must be publicly readable).